

MSS - module 2

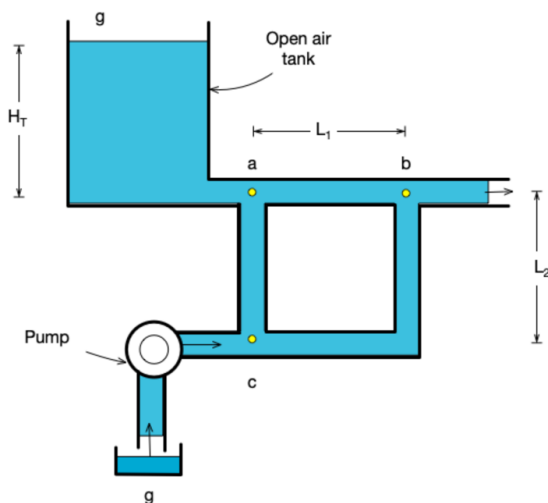
Written exam 2025/06/23

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Modellin part

Exercise 1



Compute the

1. Incidence matrix
2. node and loop equations
3. constitutive equations
4. final set of hydraulic system of equations

```
clear all
```

```
% Definitions of through variables
```

```
syms t Q_tank(t) Q_ab(t) Q_ac(t) Q_cb(t) Q_out(t) Q_pump(t)
```

```
tau = [Q_tank(t) ; Q_ab(t) ; Q_ac(t) ; Q_cb(t) ; Q_out(t) ; Q_pump(t)]
```

tau =

$$\begin{pmatrix} Q_{\text{tank}}(t) \\ Q_{\text{ab}}(t) \\ Q_{\text{ac}}(t) \\ Q_{\text{cb}}(t) \\ Q_{\text{out}}(t) \\ Q_{\text{pump}}(t) \end{pmatrix}$$

% Branch through coordinates

tau_b = [Q_tank(t) ;Q_ab(t) ;Q_ac(t)]

tau_b =

$$\begin{pmatrix} Q_{\text{tank}}(t) \\ Q_{\text{ab}}(t) \\ Q_{\text{ac}}(t) \end{pmatrix}$$

% Chord through coordinates

tau_c =[Q_cb(t) ;Q_out(t) ;Q_pump(t)]

tau_c =

$$\begin{pmatrix} Q_{\text{cb}}(t) \\ Q_{\text{out}}(t) \\ Q_{\text{pump}}(t) \end{pmatrix}$$

%definition of accross variables

syms P_tank(t) P_ab(t) P_ac(t) P_cb(t) P_out(t) P_pump(t)

alpha = [P_tank(t) ;P_ab(t) ;P_ac(t); P_cb(t); P_out(t); P_pump(t)]

alpha =

$$\begin{pmatrix} P_{\text{tank}}(t) \\ P_{\text{ab}}(t) \\ P_{\text{ac}}(t) \\ P_{\text{cb}}(t) \\ P_{\text{out}}(t) \\ P_{\text{pump}}(t) \end{pmatrix}$$

% Branch across coordinates

alpha_b= [P_tank(t) ;P_ab(t) ;P_ac(t)]

alpha_b =

$$\begin{pmatrix} P_{\text{tank}}(t) \\ P_{\text{ab}}(t) \\ P_{\text{ac}}(t) \end{pmatrix}$$

% Chord across coordinates

```
alpha_c=[P_cb(t) ;P_out(t) ;P_pump(t)]
```

alpha_c =

$$\begin{pmatrix} P_{\text{cb}}(t) \\ P_{\text{out}}(t) \\ P_{\text{pump}}(t) \end{pmatrix}$$

% definition of parameters

```
syms H_t L1 L2 Lout D g rho mu d
```

Derived parameters

%Resistance of pipe with circula cross sections

```
c= 128*mu/(pi*d^4);
```

```
R_ac = L2*c;
```

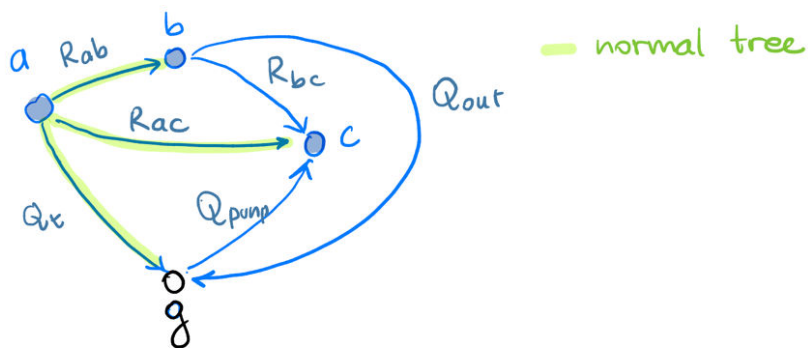
```
R_ab = L1*c;
```

```
R_bc = (L1+L2)*c;
```

```
%R_out = Lout*c;
```

```
C = pi*(D/2)^2/rho/g;
```

The linear graph and the corresponding Normal tree in the following picture. The normal tree brunch are highlighted in green



Reduced Incidence matrix

```
% Q_tank(t) ;Q_ab(t) ;Q_ac(t); Q_cb(t); Q_out(t); Q_pump(t)
```

```
A=[-1 -1 -1 0 0 0;...
```

```
0 1 0 1 -1 0;...
```

$$\begin{bmatrix} 0 & 0 & 1 & -1 & 0 & 1 \end{bmatrix}$$

A = 3x6

$$\begin{bmatrix} -1 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{bmatrix}$$

Echelon form of reduced incidence matrix

$$A_f = \text{rref}(A) \% [I_b \mid A_c]$$

Af = 3x6

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{bmatrix}$$

$$A_c = A_f(1:3, 4:6)$$

Ac = 3x3

$$\begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Orthogonal matrix

$$B_b = -A_c.' \% [A_b, I_c]$$

Bb = 3x3

$$\begin{bmatrix} 0 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

Loop equations

$$\text{loop_eqns} = \alpha_c == -B_b * \alpha_b$$

loop_eqns =

$$\begin{pmatrix} P_{cb}(t) = P_{ab}(t) - P_{ac}(t) \\ P_{out}(t) = P_{tank}(t) - P_{ab}(t) \\ P_{pump}(t) = P_{ac}(t) - P_{tank}(t) \end{pmatrix}$$

Node equations

$$\text{node_eqns} = \tau_b == -A_c * \tau_c$$

node_eqns =

$$\begin{pmatrix} Q_{tank}(t) = Q_{pump}(t) - Q_{out}(t) \\ Q_{ab}(t) = Q_{out}(t) - Q_{cb}(t) \\ Q_{ac}(t) = Q_{cb}(t) - Q_{pump}(t) \end{pmatrix}$$

Constitutive equations

```
syms f(t)

const_eqns= [
    C*diff(P_tank(t),t) == Q_tank(t);
    P_ab(t) == R_ab*Q_ab(t);
    P_ac(t) == R_ac*Q_ac(t);
    P_cb(t) == R_bc*Q_cb(t);
    Q_pump(t) == f(t)
]
```

const_eqns =

$$\begin{pmatrix} \frac{\pi D^2}{4 g \rho} \frac{\partial P_{\text{tank}}(t)}{\partial t} = Q_{\text{tank}}(t) \\ P_{\text{ab}}(t) = \frac{128 L_1 \mu Q_{\text{ab}}(t)}{d^4 \pi} \\ P_{\text{ac}}(t) = \frac{128 L_2 \mu Q_{\text{ac}}(t)}{d^4 \pi} \\ P_{\text{cb}}(t) = \frac{128 \mu Q_{\text{cb}}(t) (L_1 + L_2)}{d^4 \pi} \\ Q_{\text{pump}}(t) = f(t) \end{pmatrix}$$

```
%P_out(t) == R_out*Q_out(t);
tmp0 =
subs(const_eqns,num2cell(lhs(loop_eqns)),num2cell(rhs(loop_eqns)))
```

tmp0 =

$$\begin{pmatrix} \frac{\pi D^2}{4 g \rho} \frac{\partial P_{\text{tank}}(t)}{\partial t} = Q_{\text{tank}}(t) \\ P_{\text{ab}}(t) = \frac{128 L_1 \mu Q_{\text{ab}}(t)}{d^4 \pi} \\ P_{\text{ac}}(t) = \frac{128 L_2 \mu Q_{\text{ac}}(t)}{d^4 \pi} \\ P_{\text{ab}}(t) - P_{\text{ac}}(t) = \frac{128 \mu Q_{\text{cb}}(t) (L_1 + L_2)}{d^4 \pi} \\ Q_{\text{pump}}(t) = f(t) \end{pmatrix}$$

```
dae= subs(tmp0,num2cell(lhs(node_eqns)),num2cell(rhs(node_eqns)))
```

dae =

$$\left(\begin{array}{l} \frac{\pi D^2 \frac{\partial}{\partial t} P_{\text{tank}}(t)}{4 g \rho} = Q_{\text{pump}}(t) - Q_{\text{out}}(t) \\ P_{\text{ab}}(t) = - \frac{128 L_1 \mu (Q_{\text{cb}}(t) - Q_{\text{out}}(t))}{d^4 \pi} \\ P_{\text{ac}}(t) = \frac{128 L_2 \mu (Q_{\text{cb}}(t) - Q_{\text{pump}}(t))}{d^4 \pi} \\ P_{\text{ab}}(t) - P_{\text{ac}}(t) = \frac{128 \mu Q_{\text{cb}}(t) (L_1 + L_2)}{d^4 \pi} \\ Q_{\text{pump}}(t) = f(t) \end{array} \right)$$

Define the vector of algebraic variables

```
alg_vars = [P_ab(t), P_ac(t), Q_cb(t), Q_pump(t)]
```

$$\text{alg_vars} = (P_{\text{ab}}(t) \ P_{\text{ac}}(t) \ Q_{\text{cb}}(t) \ Q_{\text{pump}}(t))$$

Define vector of algebraic variables for substitution

```
a = sym("a", [1 4])
```

$$\mathbf{a} = (a_1 \ a_2 \ a_3 \ a_4)$$

Solve system of algebraic variables

```
S=solve(subs([dae(2:5)],num2cell(alg_vars),num2cell(a)),a)
```

```
S = struct with fields:
  a1: (64*L1*mu*(L1*Q_out(t) + 2*L2*Q_out(t) - L2*f(t)))/(d^4*pi*(L1 + L2))
  a2: -(64*L2*mu*(2*L1*f(t) - L1*Q_out(t) + L2*f(t)))/(d^4*pi*(L1 + L2))
  a3: (L1*Q_out(t) + L2*f(t))/(2*(L1 + L2))
  a4: f(t)
```

```
[ S.a1;S.a2;S.a3;S.a4]
```

ans =

$$\left(\begin{array}{l} \frac{64 L_1 \mu (L_1 Q_{\text{out}}(t) + 2 L_2 Q_{\text{out}}(t) - L_2 f(t))}{\sigma_1} \\ - \frac{64 L_2 \mu (2 L_1 f(t) - L_1 Q_{\text{out}}(t) + L_2 f(t))}{\sigma_1} \\ \frac{L_1 Q_{\text{out}}(t) + L_2 f(t)}{2 (L_1 + L_2)} \\ f(t) \end{array} \right)$$

where

$$\sigma_1 = d^4 \pi (L_1 + L_2)$$

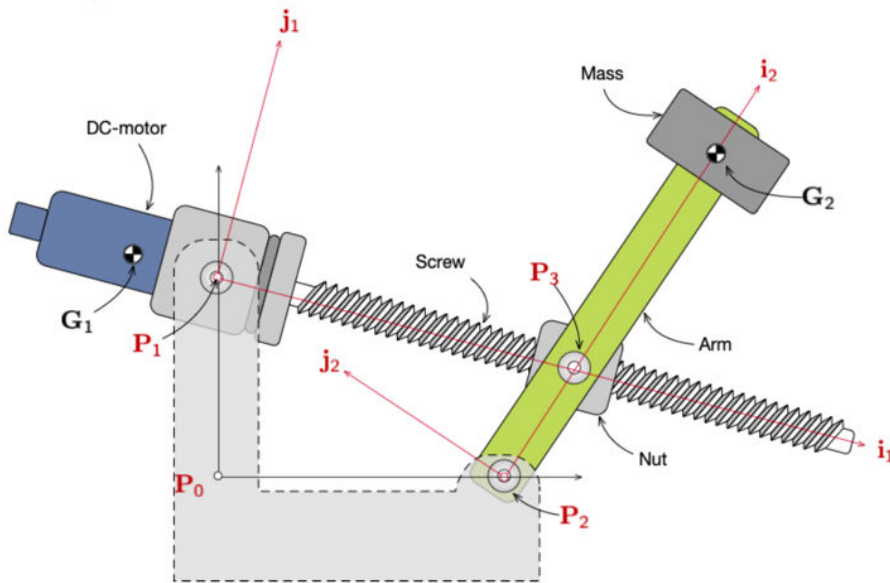
Substitute in the remaining equation and obtain the ode. The pressure (or height) of the tank is a function of the pump flow rate and the output flow rate which are two given functions

```
ode= subs(dae(1),num2cell(a),num2cell([ S.a1 S.a2 S.a3 S.a4]))
```

ode =

$$\frac{\pi D^2 \frac{\partial}{\partial t} P_{\text{tank}}(t)}{4 g \rho} = Q_{\text{pump}}(t) - Q_{\text{out}}(t)$$

Exercise 2



Compute the

1. LG + normal tree
2. T-Graph and loop equations
3. Node equations
4. Newton Euler equations

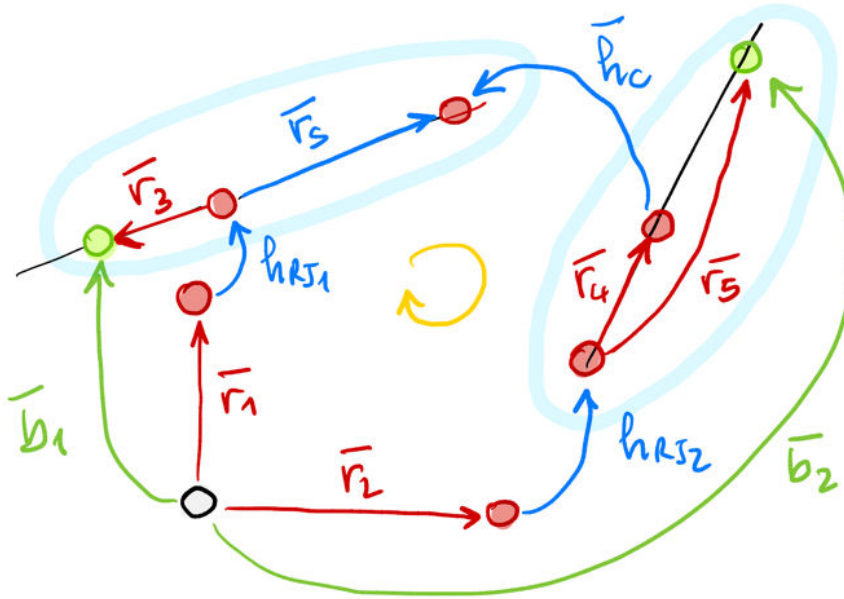
```
clear all
clear;
addpath('MBsLib_v2_1/')
```

Definition of variables and parameters

```
syms theta_1(t) theta_2(t) s(t) theta_a(t)

% Parameters
syms L H x_G1 x_G2 L3 tau_s
```

Linear Graph and normal tree



The normal tree contains the joint: {revolute joint RJ1, revolute joint RJ2 and screw joint rs}. The corresponding branch coordinates are

$$\mathbf{q} = \{\theta_1(t), \theta_2(t), s(t)\}$$

The joint h_c is a revolute joint. The combination of screw plus revolute is a cam joint.

We next define the joint matrices and the vectors (edges) of the T-graph

Rotation matrix of first revolute joint and definition of vectors

```
RF1 = rotate('Z', theta_1(t))
```

RF1 =

$$\begin{pmatrix} \cos(\theta_1(t)) & -\sin(\theta_1(t)) & 0 & 0 \\ \sin(\theta_1(t)) & \cos(\theta_1(t)) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Revolute joint translation in P1

```
hRJ1 = [0;0;0;0];
```

```
% Position in world frame of revolute joint in P1
```

```
r1 = [0;H;0;0];
```

```
% Position in world frame of revolute joint in P2
```

```
r2 = [L;0;0;0];
```

Local vector of body 1 centre of mass in world coordinates

$$r3 = -RF1*[-x_{G1};0;0;0]$$

$$r3 =$$

$$\begin{pmatrix} x_{G1} \cos(\theta_1(t)) \\ x_{G1} \sin(\theta_1(t)) \\ 0 \\ 0 \end{pmatrix}$$

Translation of screw joint

```
% Screw translation as a function of screw rotation theta_a(t) about x-axis
of RF1
s(t) == tau_s*theta_a(t)
```

$$ans = s(t) = \tau_s \theta_a(t)$$

Vector of translation of screw joint

$$rs = RF1*[s(t);0;0;0]$$

$$rs =$$

$$\begin{pmatrix} \cos(\theta_1(t)) s(t) \\ \sin(\theta_1(t)) s(t) \\ 0 \\ 0 \end{pmatrix}$$

Rotation matrix of second revolute joint and definition of vectors

$$RF2 = \text{rotate}('Z', \theta_2(t))$$

$$RF2 =$$

$$\begin{pmatrix} \cos(\theta_2(t)) & -\sin(\theta_2(t)) & 0 & 0 \\ \sin(\theta_2(t)) & \cos(\theta_2(t)) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Revolute joint translation in P2

$$hRJ2 = [0;0;0;0];$$

Local vector of body 2 centre of mass in world coordinates

$$r4 = RF2*[x_{G2};0;0;0]$$

$$r4 =$$

$$\begin{pmatrix} x_{G2} \cos(\theta_2(t)) \\ x_{G2} \sin(\theta_2(t)) \\ 0 \\ 0 \end{pmatrix}$$

Vector of position of revolute joint in P3

$$r5 = RF2*[L3;0;0;0]$$

$$r5 =$$

$$\begin{pmatrix} L_3 \cos(\theta_2(t)) \\ L_3 \sin(\theta_2(t)) \\ 0 \\ 0 \end{pmatrix}$$

Revolute joint translation in P3

$$hRJ3 = [0;0;0;0]; \% hc$$

T-graph: loop equations

There are three loop equations: one for the mechanical joints that links the branch coordinates

$$\bar{r}_1 + \bar{h}_{RJ1} + \bar{r}_5 - \bar{h}_c - \bar{r}_4 - \bar{h}_{RJ2} - \bar{r}_2 = 0$$

$$\bar{b}_1 = \bar{r}_1 + \bar{h}_{RJ1} + \bar{r}_5$$

$$\bar{b}_2 = \bar{r}_2 + \bar{h}_{RJ2} + \bar{r}_5$$

The first loop equation (the cam joint equation) is projected in the reaction and motion space of the cam joint

$$eqL1 = r1+hRJ1+rs-hRJ3-r4-hRJ2-r2$$

$$eqL1 =$$

$$\begin{pmatrix} \cos(\theta_1(t)) s(t) - x_{G2} \cos(\theta_2(t)) - L \\ H - x_{G2} \sin(\theta_2(t)) + \sin(\theta_1(t)) s(t) \\ 0 \\ 0 \end{pmatrix}$$

% Unit vectors of the reaction and motion space of the cam joint

$$ic = RF1(:,1)$$

$$ic =$$

$$\begin{pmatrix} \cos(\theta_1(t)) \\ \sin(\theta_1(t)) \\ 0 \\ 0 \end{pmatrix}$$

$$jc = RF1(:,2)$$

$$j c = \begin{pmatrix} -\sin(\theta_1(t)) \\ \cos(\theta_1(t)) \\ 0 \\ 0 \end{pmatrix}$$

Constraint equations

```
Phi = [simplify(expand(eqL1.*ic==0));
       combine(expand(eqL1.*jc==0))]
```

$$\text{Phi} = \begin{pmatrix} s(t) + H \sin(\theta_1(t)) = x_{G2} \cos(\theta_1(t) - \theta_2(t)) + L \cos(\theta_1(t)) \\ L \sin(\theta_1(t)) + H \cos(\theta_1(t)) - x_{G2} \cos(\theta_1(t)) \sin(\theta_2(t)) + x_{G2} \cos(\theta_2(t)) \sin(\theta_1(t)) = 0 \end{pmatrix}$$

The other two loop equations are used to compute the position of bodies' CoM

```
% Position, velocity and acceleration of body 1
b1 = r1+hRJ1+r3
```

$$b1 = \begin{pmatrix} x_{G1} \cos(\theta_1(t)) \\ H + x_{G1} \sin(\theta_1(t)) \\ 0 \\ 0 \end{pmatrix}$$

```
V_b1 = diff(b1,t)
```

$$V_b1 = \begin{pmatrix} -x_{G1} \sin(\theta_1(t)) \frac{\partial}{\partial t} \theta_1(t) \\ x_{G1} \cos(\theta_1(t)) \frac{\partial}{\partial t} \theta_1(t) \\ 0 \\ 0 \end{pmatrix}$$

```
A_b1 = diff(b1,t,t)
```

$$A_b1 = \begin{pmatrix} -x_{G1} \sin(\theta_1(t)) \frac{\partial^2}{\partial t^2} \theta_1(t) - x_{G1} \cos(\theta_1(t)) \left(\frac{\partial}{\partial t} \theta_1(t) \right)^2 \\ x_{G1} \cos(\theta_1(t)) \frac{\partial^2}{\partial t^2} \theta_1(t) - x_{G1} \sin(\theta_1(t)) \left(\frac{\partial}{\partial t} \theta_1(t) \right)^2 \\ 0 \\ 0 \end{pmatrix}$$

% Position, velocity and acceleration of body 2

b2 = r2+hRJ2+r5

b2 =

$$\begin{pmatrix} L + L_3 \cos(\theta_2(t)) \\ L_3 \sin(\theta_2(t)) \\ 0 \\ 0 \end{pmatrix}$$

V_b2 = diff(b2,t)

V_b2 =

$$\begin{pmatrix} -L_3 \sin(\theta_2(t)) \frac{\partial}{\partial t} \theta_2(t) \\ L_3 \cos(\theta_2(t)) \frac{\partial}{\partial t} \theta_2(t) \\ 0 \\ 0 \end{pmatrix}$$

A_b2 = diff(b2,t,t)

A_b2 =

$$\begin{pmatrix} -L_3 \sin(\theta_2(t)) \frac{\partial^2}{\partial t^2} \theta_2(t) - L_3 \cos(\theta_2(t)) \left(\frac{\partial}{\partial t} \theta_2(t) \right)^2 \\ L_3 \cos(\theta_2(t)) \frac{\partial^2}{\partial t^2} \theta_2(t) - L_3 \sin(\theta_2(t)) \left(\frac{\partial}{\partial t} \theta_2(t) \right)^2 \\ 0 \\ 0 \end{pmatrix}$$

R-graph: loop equations

Are not necessary since they are trivial in this case

Node equations

We do not consider the weight force but it can easily added.

We define the reaction forces:

```
syms fx_j1(t) fy_j1(t) fx_j2(t) fy_j2(t) n(t) fs(t) Ta(t)
```

```
FRJ1 = [fx_j1(t);fy_j1(t);0;0] % revolute joint 1 reaction force in world frame
```

FRJ1 =

$$\begin{pmatrix} f_{xj1}(t) \\ f_{yj1}(t) \\ 0 \\ 0 \end{pmatrix}$$

```
FRJ2 = [fx_j2(t);fy_j2(t);0;0] % revolute joint 2 reaction force in world frame
```

FRJ2 =

$$\begin{pmatrix} f_{xj2}(t) \\ f_{yj2}(t) \\ 0 \\ 0 \end{pmatrix}$$

The cam joint is special. It should have a reaction force in the normal direction to the screw, thus it has to be defined in RF1 frame

However there is also a force to control the screw position which is along the screw motion space j_c . We combine them in one single force FS

```
FS_b = [fs(t);n(t);0;0]; % cam + screw reaction force in body frame
FS = RF1*FS_b % in world frame
```

FS =

$$\begin{pmatrix} \cos(\theta_1(t)) fs(t) - \sin(\theta_1(t)) n(t) \\ \cos(\theta_1(t)) n(t) + \sin(\theta_1(t)) fs(t) \\ 0 \\ 0 \end{pmatrix}$$

```
% FS = RF1*[fs(t);n(t);0;0] % cam + screw reaction force
dyn_eq_screw = fs(t) == Ta(t)/tau_s % Ta(t) is the screw actuation torque
```

dyn_eq_screw =

$$fs(t) = \frac{Ta(t)}{\tau_s}$$

Node equations

```
Fb1 = FRJ1+FS
```

Fb1 =

$$\begin{pmatrix} f_{xj1}(t) + \cos(\theta_1(t)) fs(t) - \sin(\theta_1(t)) n(t) \\ f_{yj1}(t) + \cos(\theta_1(t)) n(t) + \sin(\theta_1(t)) fs(t) \\ 0 \\ 0 \end{pmatrix}$$

```
Fb2 = FRJ2-FS
```

Fb2 =

$$\begin{pmatrix} f_{xj2}(t) - \cos(\theta_1(t)) f_s(t) + \sin(\theta_1(t)) n(t) \\ f_{yj2}(t) - \cos(\theta_1(t)) n(t) - \sin(\theta_1(t)) f_s(t) \\ 0 \\ 0 \end{pmatrix}$$

Compute and project moment in body frame

```
FRJ1_b = inv_frame(RF1)*FRJ1; % in body frame
Tb1 = simplify(RF1(1:3,1:3).*(skew_matrix(-r3(1:3))*FRJ1(1:3)+skew_matrix(-r3(1:3)+rs(1:3))*FS(1:3)))
```

Tb1 =

$$\begin{pmatrix} 0 \\ 0 \\ n(t) s(t) - x_{G1} n(t) + x_{G1} \sin(\theta_1(t)) f_{xj1}(t) - x_{G1} \cos(\theta_1(t)) f_{yj1}(t) \end{pmatrix}$$

```
Tb2 = simplify(RF2(1:3,1:3).*(skew_matrix(-r5(1:3))*FRJ2(1:3)-skew_matrix(r5(1:3)-r4(1:3))*FS(1:3)))
```

Tb2 =

$$\begin{pmatrix} 0 \\ 0 \\ \sin(\theta_2(t)) (\cos(\theta_1(t)) f_s(t) - \sin(\theta_1(t)) n(t)) (L_3 - x_{G2}) - \cos(\theta_2(t)) (\cos(\theta_1(t)) n(t) + \sin(\theta_1(t)) f_s(t)) (L_3 - \end{pmatrix}$$

Newton Euler equations (constitutive equations)

```
syms m1 m2 Iz1 Iz2
```

```
i0 = [1;0;0;0];
j0 = [0;1;0;0];
k0 = [0;0;1;0];
```

```
% Newton Euler first body
```

```
tmp_N = m1*A_b1 == Fb1;
```

```
[tmp_N.*i0;
 tmp_N.*j0;
 Iz1*diff(theta_1(t),t,t)==simplify(Tb1.*k0(1:3))
]
```

ans =

$$\begin{pmatrix} -m_1 (x_{G1} \sin(\theta_1(t)) \sigma_1 + x_{G1} \cos(\theta_1(t)) \sigma_2) = f_{x_{j1}}(t) + \cos(\theta_1(t)) f_s(t) - \sin(\theta_1(t)) n(t) \\ -m_1 (x_{G1} \sin(\theta_1(t)) \sigma_2 - x_{G1} \cos(\theta_1(t)) \sigma_1) = f_{y_{j1}}(t) + \cos(\theta_1(t)) n(t) + \sin(\theta_1(t)) f_s(t) \\ I_{z1} \sigma_1 = n(t) s(t) - x_{G1} n(t) + x_{G1} \sin(\theta_1(t)) f_{x_{j1}}(t) - x_{G1} \cos(\theta_1(t)) f_{y_{j1}}(t) \end{pmatrix}$$

where

$$\sigma_1 = \frac{\partial^2}{\partial t^2} \theta_1(t)$$

$$\sigma_2 = \left(\frac{\partial}{\partial t} \theta_1(t) \right)^2$$

% Newton Euler second body

tmp_N = m2*A_b2 == Fb2;

```
[tmp_N.*i0;
 tmp_N.*j0;
 Iz2*diff(theta_2(t),t,t)==simplify(Tb2.*k0(1:3))
]
```

ans =

$$\begin{pmatrix} -m_2 (L_3 \sin(\theta_2(t)) \sigma_1 + L_3 \cos(\theta_2(t)) \sigma_6) = f_{x_{j2}}(t) - \sigma_5 + \sigma_2 \\ -m_2 (L_3 \sin(\theta_2(t)) \sigma_6 - L_3 \cos(\theta_2(t)) \sigma_1) = f_{y_{j2}}(t) - \sigma_4 - \sigma_3 \\ I_{z2} \sigma_1 = \sin(\theta_2(t)) (\sigma_5 - \sigma_2) (L_3 - x_{G2}) - \cos(\theta_2(t)) (\sigma_4 + \sigma_3) (L_3 - x_{G2}) - L_3 \cos(\theta_2(t)) f_{y_{j2}}(t) + L_3 \sin(\theta_2(t)) f_{x_{j2}}(t) \end{pmatrix}$$

where

$$\sigma_1 = \frac{\partial^2}{\partial t^2} \theta_2(t)$$

$$\sigma_2 = \sin(\theta_1(t)) n(t)$$

$$\sigma_3 = \sin(\theta_1(t)) f_s(t)$$

$$\sigma_4 = \cos(\theta_1(t)) n(t)$$

$$\sigma_5 = \cos(\theta_1(t)) f_s(t)$$

$$\sigma_6 = \left(\frac{\partial}{\partial t} \theta_2(t) \right)^2$$